

**GOLD COAST SEAWAY, Australia**WMO: **945800**Lat: **27.9388S**Lon: **153.4283E**Elev: **14**StdP: **14.69**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 49.3       | 51.4 | 30.0                          | 24.1 | 61.6 | 33.2 | 27.7 | 60.8 | 28.8                  | 65.6 | 25.7 | 65.2 | 7.5                   | 290  | 0.452 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.0                   | 86.9            | 74.1 | 84.9 | 73.8 | 83.3 | 73.2 | 78.3                | 82.2 | 77.0 | 81.0 | 75.9 | 80.1 | 14.9                 | 30   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.3                            | 142.2 | 80.1 | 75.9 | 135.6 | 79.1 | 74.6 | 129.6 | 78.3 | 41.9          | 82.5 | 40.5 | 80.9 | 39.4 | 80.3 | 84.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |             |             |             |             |             |              |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years  |             | n=20 years  |             | n=50 years  |              |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min         | Max         | Min         | Max         | Min         | Max          | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)         | (l)         | (m)         | (n)         | (o)         | (p)          |    |    |
| (4) <b>28.8</b>   | <b>25.5</b> | <b>22.8</b> | <b>44.8</b>                | <b>93.4</b> | <b>2.8</b>         | <b>3.3</b> | <b>42.8</b>                                        | <b>95.7</b> | <b>41.2</b> | <b>97.6</b> | <b>39.7</b> | <b>99.4</b> | <b>37.7</b> | <b>101.8</b> |    |    |
| (5)               |             |             | <b>39.9</b>                | <b>80.0</b> | <b>2.2</b>         | <b>1.6</b> | <b>38.3</b>                                        | <b>81.2</b> | <b>37.0</b> | <b>82.1</b> | <b>35.7</b> | <b>82.9</b> | <b>34.1</b> | <b>84.1</b>  |    |    |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 69.8   | 77.1  | 76.8  | 75.3  | 71.4  | 66.7  | 62.9  | 61.5  | 62.6  | 66.4  | 69.4  | 72.6  | 75.0  |
|                                                                               | DBStd       | 6.30   | 2.83  | 2.90  | 2.80  | 2.74  | 3.11  | 3.26  | 3.09  | 3.22  | 3.39  | 3.42  | 3.31  | 2.97  |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | HDD65       | 327    | 0     | 0     | 0     | 1     | 19    | 78    | 115   | 87    | 22    | 5     | 0     | 0     |
|                                                                               | CDD50       | 7219   | 841   | 751   | 786   | 642   | 517   | 386   | 356   | 392   | 493   | 601   | 678   | 776   |
|                                                                               | CDD65       | 2071   | 376   | 331   | 321   | 193   | 71    | 14    | 6     | 14    | 65    | 140   | 229   | 311   |
|                                                                               | CDH74       | 10134  | 2487  | 2162  | 1752  | 616   | 105   | 10    | 9     | 39    | 156   | 400   | 843   | 1555  |
|                                                                               | CDH80       | 1519   | 449   | 392   | 232   | 45    | 1     | 0     | 0     | 3     | 14    | 44    | 103   | 236   |
| Wind                                                                          | WSAvg       | 11.3   | 12.6  | 12.8  | 12.5  | 11.3  | 10.0  | 9.9   | 9.7   | 10.2  | 11.0  | 11.8  | 11.6  | 11.8  |
| Precipitation                                                                 | PrecAvg     | 59.9   | 6.3   | 7.1   | 7.1   | 5.6   | 5.8   | 3.3   | 3.1   | 2.5   | 1.8   | 4.1   | 4.9   | 5.9   |
|                                                                               | PrecMax     | 105.1  | 30.6  | 22.6  | 20.5  | 25.2  | 21.1  | 27.2  | 16.0  | 9.0   | 7.5   | 20.3  | 16.3  | 13.1  |
|                                                                               | PrecMin     | 33.5   | 0.4   | 0.0   | 0.4   | 0.0   | 0.7   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.7   | 0.0   |
|                                                                               | PrecStd     | 17.5   | 5.2   | 5.6   | 4.9   | 5.5   | 4.8   | 4.9   | 3.1   | 2.2   | 1.6   | 3.5   | 3.2   | 3.4   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 89.6  | 89.2  | 87.9  | 84.0  | 79.2  | 75.3  | 74.8  | 78.0  | 81.5  | 84.7  | 86.2  | 88.7  |
|                                                                               |             | MCWB   | 76.5  | 75.4  | 74.5  | 70.9  | 66.4  | 63.4  | 61.6  | 61.4  | 62.8  | 65.6  | 68.9  | 71.3  |
|                                                                               | 2%          | DB     | 86.1  | 86.1  | 84.6  | 81.0  | 76.8  | 72.7  | 72.0  | 74.4  | 77.7  | 80.4  | 82.6  | 84.6  |
|                                                                               |             | MCWB   | 75.2  | 75.9  | 74.3  | 69.5  | 65.5  | 62.6  | 60.4  | 61.0  | 63.9  | 67.8  | 69.9  | 72.8  |
|                                                                               | 5%          | DB     | 84.2  | 84.2  | 82.5  | 79.0  | 74.8  | 70.9  | 70.1  | 72.0  | 75.4  | 78.1  | 80.5  | 82.5  |
|                                                                               |             | MCWB   | 74.6  | 74.7  | 73.3  | 69.1  | 64.8  | 61.6  | 60.1  | 60.2  | 64.2  | 67.4  | 70.0  | 72.3  |
|                                                                               | 10%         | DB     | 82.5  | 82.5  | 80.7  | 77.4  | 73.2  | 69.3  | 68.4  | 69.9  | 73.5  | 75.9  | 78.6  | 80.7  |
|                                                                               |             | MCWB   | 73.7  | 73.7  | 72.2  | 68.5  | 64.7  | 60.8  | 59.1  | 59.4  | 63.3  | 66.6  | 69.5  | 71.7  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 79.4  | 80.2  | 79.8  | 74.3  | 71.9  | 68.2  | 67.4  | 67.2  | 69.6  | 72.9  | 75.2  | 77.8  |
|                                                                               |             | MCDB   | 83.0  | 83.8  | 82.9  | 78.7  | 73.9  | 70.9  | 70.2  | 71.4  | 74.0  | 77.1  | 79.3  | 81.2  |
|                                                                               | 2%          | WB     | 77.9  | 78.4  | 77.6  | 72.6  | 70.0  | 66.0  | 65.2  | 65.2  | 68.1  | 71.1  | 73.6  | 76.1  |
|                                                                               |             | MCDB   | 82.3  | 82.5  | 80.6  | 77.7  | 73.3  | 69.5  | 68.4  | 70.1  | 73.3  | 75.9  | 77.7  | 79.9  |
|                                                                               | 5%          | WB     | 76.5  | 76.8  | 76.0  | 71.2  | 68.1  | 64.5  | 63.2  | 63.4  | 66.8  | 69.7  | 72.5  | 74.9  |
|                                                                               |             | MCDB   | 81.2  | 81.0  | 79.5  | 76.9  | 72.5  | 68.2  | 67.2  | 68.9  | 72.2  | 74.2  | 77.1  | 79.3  |
|                                                                               | 10%         | WB     | 75.5  | 75.6  | 74.4  | 70.0  | 66.3  | 63.1  | 61.3  | 61.9  | 65.6  | 68.4  | 71.4  | 73.7  |
|                                                                               |             | MCDB   | 80.2  | 80.1  | 78.6  | 75.8  | 71.1  | 67.2  | 66.0  | 67.6  | 71.0  | 73.6  | 76.5  | 78.3  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.0  | 10.3  | 10.6  | 12.0  | 13.1  | 12.7  | 14.6  | 15.3  | 14.3  | 12.5  | 11.2  | 10.7  |
|                                                                               |             | MCDBR  | 12.9  | 13.2  | 13.4  | 14.6  | 15.4  | 14.6  | 16.8  | 18.5  | 17.6  | 16.4  | 14.2  | 13.9  |
|                                                                               |             | MCWBR  | 5.7   | 5.9   | 6.2   | 7.0   | 7.7   | 7.8   | 8.9   | 9.1   | 8.7   | 8.1   | 6.4   | 6.2   |
|                                                                               |             | MCWBR  | 10.9  | 11.0  | 11.2  | 12.4  | 11.7  | 10.9  | 12.6  | 14.8  | 13.9  | 13.2  | 11.9  | 11.5  |
|                                                                               | 5% WB       | MCWBR  | 5.8   | 5.9   | 6.3   | 6.6   | 7.0   | 7.1   | 8.1   | 8.6   | 7.8   | 7.6   | 6.2   | 5.9   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.390 | 0.377 | 0.360 | 0.340 | 0.320 | 0.315 | 0.305 | 0.317 | 0.360 | 0.384 | 0.382 | 0.396 |
|                                                                               | taud        |        | 2.504 | 2.551 | 2.595 | 2.632 | 2.638 | 2.647 | 2.660 | 2.605 | 2.479 | 2.441 | 2.486 | 2.472 |
|                                                                               | Ebn at Noon |        | 302   | 301   | 297   | 288   | 280   | 275   | 283   | 291   | 292   | 295   | 302   | 301   |
|                                                                               | Edh at Noon |        | 36    | 34    | 31    | 27    | 24    | 23    | 23    | 27    | 34    | 37    | 37    | 38    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 2056  | 1879  | 1611  | 1362  | 1131  | 933   | 1090  | 1371  | 1695  | 1891  | 2066  | 2053  |
|                                                                               | RadStd      |        | 163   | 150   | 93    | 68    | 62    | 63    | 85    | 103   | 133   | 162   | 161   | 213   |